

## Segment 1(Electrochemistry)

**Conductance:** Conductance is the ability of a material or solution to allow the flow of electric current. It is the reciprocal of resistance (R). (Aut-24)

**Specific Conductance:** Specific conductance ( $\kappa$ ) is the conductance of 1 cm<sup>3</sup> of solution placed between two electrodes 1 cm apart.

**Molar Conductance:** Molar conductance ( $\Lambda_m$ ) is the conductance of all ions produced from one mole of electrolyte when dissolved in solution. (Aut-24)

**Ionization:** Formation of ions from neutral molecules (e.g.,  $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ ).

**Dissociation:** Separation of ions that already exist in a compound (e.g.,  $\text{NaCl(s)} \rightarrow \text{Na}^+ + \text{Cl}^-$  in water).

**Transport Number:** Transport number (t) is the fraction of total electric current carried by a particular ion.  
 $t^+ + t^- = 1$  (Aut-24)

### Difference Between Ionization and Dissociation(Aut-24)

| Ionization                                                  | Dissociation                                        |
|-------------------------------------------------------------|-----------------------------------------------------|
| Formation of ions from neutral molecules                    | Separation of already existing ions in a compound.  |
| Mainly in covalent compounds.                               | Mainly in ionic compounds.                          |
| $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$           | $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$ |
| Involves chemical change (bond breaking and ion formation). | Physical process (ions separate in solvent).        |
| Requires more energy for bond breaking.                     | Requires less energy (just separation in water).    |
| Slow process.                                               | Fast process.                                       |

## Difference Between Electrolytes and Non-Electrolytes

| Electrolytes                                                     | Non-Electrolytes                            |
|------------------------------------------------------------------|---------------------------------------------|
| Substances that conduct electricity in solution or molten state. | Substances that do not conduct electricity. |
| Produce ions in solution.                                        | Do not produce ions in solution.            |
| Good or partial conductor in solution.                           | No conductivity in solution.                |
| NaCl, H <sub>2</sub> SO <sub>4</sub> , KOH                       | Sugar, Urea, Alcohol                        |
| Ionic compounds or acids/bases.                                  | Covalent compounds.                         |
| Light the bulb when dissolved in water.                          | Does not light the bulb.                    |

## Explain briefly Faraday's First Law of Electrolysis & show that $Q = Z \cdot I \cdot t$ .

(Aut-24)

### Statement:

The mass of a substance deposited or liberated at an electrode during electrolysis is directly proportional to the quantity of electricity (charge) passed through the electrolyte.

$$m \propto Q \quad \text{or} \quad m = ZQ$$

Where:

- $m$  = mass of the substance deposited (in grams)
- $Q$  = total electric charge passed (in coulombs)
- $Z$  = electrochemical equivalent of the substance

Also, we know:

$$Q = I \cdot t$$

Where:

- $I$  = current in amperes
- $t$  = time in seconds

So,

$$m = Z \cdot Q = Z \cdot (I \cdot t)$$

Therefore:

$$Q = \frac{m}{Z}$$

Prove that  $t^- = 1 / (1 + r)$ , where  $t^-$  is the transport number of anion and  $r$  is the speed ratio of the ions. (Aut-24)

$$\text{the transport number of cation} = \frac{v_+}{v_+ + v_-}$$

$$\text{the transport number of anion} = \frac{v_-}{v_+ + v_-}$$

The transport number of the cation is represented by  $t$  and that of the anion by  $t_-$ .

Thus 
$$t_+ = \frac{v_+}{v_+ + v_-} \quad \text{and} \quad t_- = \frac{v_-}{v_+ + v_-}$$

or 
$$\frac{t_+}{t_-} = \frac{v_+}{v_-} \quad \text{and} \quad t_+ + t_- = 1$$

If the speed ratio  $v_+/v_-$  be denoted by  $r$ , we have,

$$r = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+}$$

and 
$$t_- = \frac{1}{1 + r}$$

Calculate the transport number of the  $\text{Cu}^{2+}$  and Sulphate ion

**SOLVED PROBLEM 2.** In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the  $\text{Cu}^{2+}$  and  $\text{SO}_4^{2-}$  ions.

**SOLUTION**

Wt. of copper in the anode liquid before electrolysis = 0.79 g

Wt. of copper in the anode liquid after electrolysis = 0.91 g

Increase in weight = 0.91 – 0.79

= 0.12 g

Increase in weight of copper cathode in the coulometer = 0.153 g

This means that if no copper had migrated from the anode, increase in weight would have been 0.153 g.

But actual increase = 0.12

Fall in concentration due to migration of  $\text{Cu}^{2+}$  ions = 0.153 – 0.12

= 0.033

$\therefore$  Transport number of  $\text{Cu}^{2+}$  ion = 
$$\frac{0.033}{0.153} = 0.215$$

and Transport number of  $\text{SO}_4^{2-}$  ion = 
$$(1 - 0.215) = 0.785$$

## Segment 2(Colloids & Surface Chemistry)

**Colloids:** Colloids are heterogeneous systems where the particle size of the dispersed phase ranges from **1 nm to 1000 nm**, and they are uniformly distributed in a dispersion medium. (Aut-24)

**Dispersed Phase:** The substance that is distributed as fine particles in a colloidal system is called the dispersed phase.

Example: In a milk colloid, fat globules are the dispersed phase.

**Dispersion Medium:** The medium in which the dispersed phase is distributed is called the dispersion medium.

Example: In milk, water acts as the dispersion medium.

**Adsorption:** Adsorption is the process in which molecules from a gas or liquid accumulate on the surface of a solid or liquid.

**Desorption:** Desorption is the reverse process of adsorption, where adsorbed molecules leave the surface.

**Adsorption Isotherm:** It is a curve showing the variation of adsorption with pressure (or concentration) at constant temperature.

**Mention four colloids with dispersed phase and dispersion medium.**

| Colloid | Dispersed Phase | Dispersion Medium |
|---------|-----------------|-------------------|
| Milk    | Fat             | Water             |
| Fog     | Liquid water    | Air               |
| Smoke   | Solid carbon    | Air               |
| Butter  | Water           | Fat               |

**Difference Between Physical Adsorption & Chemical Adsorption**

| Physical Adsorption                                         | Chemical Adsorption                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------|
| Weak Van der Waals forces                                   | Strong chemical bonds                                                   |
| Low (20–40 kJ/mol)                                          | High (40–400 kJ/mol)                                                    |
| Reversible                                                  | Irreversible                                                            |
| Favored by low temperature                                  | Favored by high temperature                                             |
| Non-specific; any gas can be adsorbed on any solid surface. | Highly specific; depends on chemical nature of adsorbent and adsorbate. |
| Adsorption of N <sub>2</sub> or O <sub>2</sub> on charcoal  | Adsorption of H <sub>2</sub> on Ni (catalytic surface)                  |

## Difference Between Lyophilic Sol and Lyophobic Sol

| Lyophilic Sol                                                    | Lyophobic Sol                                                      |
|------------------------------------------------------------------|--------------------------------------------------------------------|
| Solvent-loving colloids                                          | Solvent-hating colloids                                            |
| Strong attraction between dispersed phase and dispersion medium. | Very weak attraction between dispersed phase and dispersion medium |
| Highly stable even on standing for a long time.                  | Unstable; easily precipitates without stabilizers.                 |
| Reversible sol                                                   | Irreversible sol                                                   |
| High viscosity due to solvation of colloidal particles.          | Viscosity is close to that of pure dispersion medium.              |
| Gum, Gelatin, Starch sol.                                        | Gold sol, Ferric hydroxide sol.                                    |

## Write assumptions of Langmuir Adsorption isotherm.

Langmuir made the following assumptions:

- 1) The layer of the gas adsorbed on the solid adsorbent is one-molecule thick.
- 2) The adsorbed layer is uniform all over the adsorbent.
- 3) There is no interaction between the adjacent adsorbed molecules.

## State Freundlich Adsorption isotherm and show graphical expression for $\log w/m$ vs $\log P$ .

Freundlich proposed an empirical relation in form of a mathematical equation

$$m/w = kP^{1/n}$$

where,

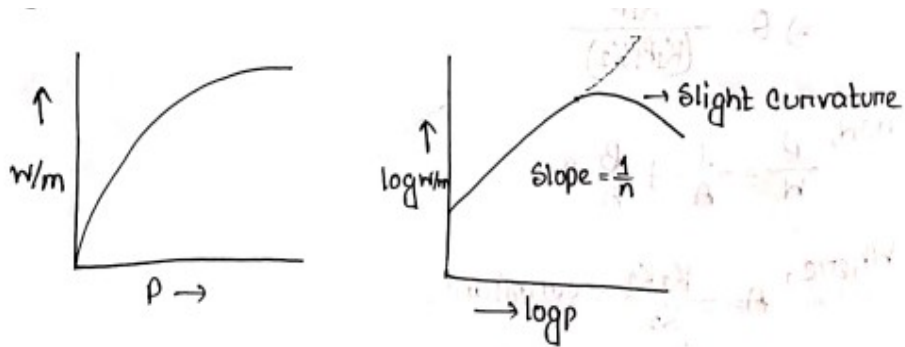
m = mass of adsorbent

w = mass of the gas absorbed

p = pressure

k & n = constant

this relation is generally represented in the form of a curve obtained by plotting the mass of the gas absorbed per unit mass of adsorbent (w/m) against equilibrium pressure.



$$\log \frac{w}{m} = \log K + \log P^{1/n}$$

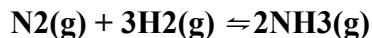
$$\Rightarrow \log \frac{w}{m} = \log K + \frac{1}{n} \log P.$$

This equation is for a straight line. Thus a plot of  $\log(w/m)$  against  $\log P$  should be a straight line with slope  $(1/n)$  and intercept  $\log K$ .

However, it is found that the plots were straight lines at a low temperature. This indicates that the Freundlich equation is approximate and does not apply to absorption of gases by solids at higher pressure.

## Segment 3(Chemical Equilibrium)

**Homogeneous Equilibrium:** If all the reactants and products are in the **same phase** (solid, liquid, or gas), the equilibrium is called homogeneous equilibrium.

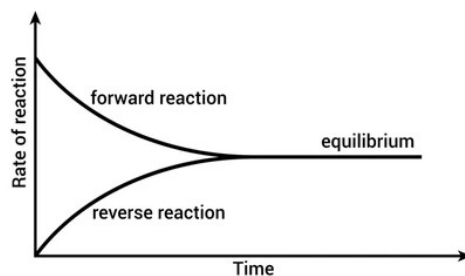


**Heterogeneous Equilibrium:** If the reactants and products are in different phases, the equilibrium is heterogeneous.



### How a chemical reaction reaches equilibrium?

In a reversible reaction, the reactants can turn into products, and vice versa. When this happens, both processes occur simultaneously: reactants change into products, and products change back into reactants. Eventually, they reach a point where the rate at which reactants become products is equal to the rate at which products turn back into reactants. This balanced state is what we call Chemical Equilibrium.



### Briefly explain Characteristics of Chemical equilibrium. (Aut-24)

There are some characteristics of Chemical Equilibrium. They are:

- 1) **Reversibility of Reaction:** Chemical Equilibrium is related only to reversible reactions. It is not applicable to irreversible reaction.
- 2) **Stability of equilibrium:** If the external conditions are not changed, the system will stay for ever.
- 3) **Easy approachability from both sides:** The equilibrium can be attained from both directions.
- 4) **Closed System:** It is only attained in closed system.
- 5) **Ineffectiveness of Catalysts:** The Catalysts have no effect on chemical equilibrium.

## Law of Mass Action and Le-Chatelier Principle.

**Law of Mass Action:** "The rate of chemical reaction is proportional to the active masses of the reactants.

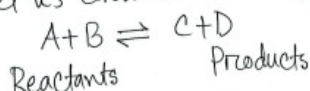
$$\text{Rate} = k[A]^a[B]^b$$

**Le-Chatelier Principle:** "If a change occurs in one of the controlling factors, such as temperature, pressure, concentration etc. under which a system is in equilibrium the system will tend to adjust itself in such a way so as to reduce the effect of that change."

### Derive Equilibrium Constant ( $K_c$ and $K_p$ ) for $A + B \rightleftharpoons C + D$ (Aut-24)

⇒ Equilibrium Constant:

Let us consider following general reaction



According to the law of mass action, the rate of forward reaction,

$$\begin{aligned} R_f &\propto [A][B] \\ R_f &= K_1 [A][B] \end{aligned}$$

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Where,  $[A]$  &  $[B]$  are the molar concentration of the reactants A & B respectively.

$K_1$  = Rate constant of the forward reaction

Similarly, the rate of backward reaction,

$$R_b \propto [C][D]$$

$$R_b = K_2 [C][D]$$

At the equilibrium, the rate of forward reaction and backward reaction remain same.

$$R_f = R_b$$

$$K_1 [A][B] = K_2 [C][D]$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C][D]}{[A][B]}$$

$$\therefore K_c = \frac{[C][D]}{[A][B]}$$

Here,  $K_c$  is called molar equilibrium constant or simply equilibrium constant of the reaction.

Where,  
 $[C]$  = Molar concentration of the product C.

$[D]$  = Molar concentration of product D.

$K_2$  = Rate constant of the backward reaction.

$$K_c = \frac{[C][D]}{[A][B]}, \quad K_p = \frac{P_C \times P_D}{P_A \times P_B}$$



Prove that  $K_p = K_c (RT)^{\Delta n}$  from  $aA + bB \rightleftharpoons cC + dD$ . (Aut-24)

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$aA + bB \rightleftharpoons cC + dD$

Forward Reaction Rate,

$$R_f \propto [A]^a \times [B]^b$$

$$\therefore R_f = k_1 [A]^a [B]^b$$

Backward Reaction Rate,

$$R_b \propto [C]^c [D]^d$$

$$\therefore R_b = k_2 [C]^c [D]^d$$

At equilibrium,

$$R_f = R_b$$

$$\text{or, } k_1 [A]^a [B]^b = k_2 [C]^c [D]^d$$

$$\text{or, } \frac{k_1}{k_2} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\text{or, } k_c = \frac{C_c \times C_d}{C_A^a \times C_B^b} \quad k_p = \frac{P_C^c \times P_D^d}{P_A^a \times P_B^b}$$

We know that,

Ideal gas equation,  $PV = nRT$

$$\text{or, } P = \frac{n}{V} RT \quad \text{or, } P = CRT \quad [C = \frac{n}{V}]$$

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Putting the value of  $P$  in the  $k_p$ ,

$$k_p = \frac{(CRT)_C^c \times (CRT)_D^d}{(CRT)_A^a \times (CRT)_B^b}$$

$$\text{or, } k_p = \frac{C_C^c \times C_D^d \times (RT)^{c+d}}{C_A^a \times C_B^b \times (RT)^{a+b}}$$

$$\text{or, } k_p = k_c \times (RT)^{(c+d)-(a+b)}$$

$$\text{or, } k_p = k_c \times RT^{(\Delta n)}$$

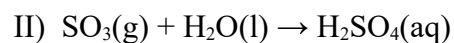
$$\therefore k_p = k_c \times (RT)^{\Delta n}$$

(Proved)

## Conditions for Maximum Sulphuric Acid Production (Contact Process) (Aut-24)

A mixture of oxygen and sulfur dioxide gas in 1:1 ratio by volume is heated at a temperature of around 450°C and at 1-2 atm pressure in presence of  $V_2O_5$  as catalyst to produce sulfur trioxide.

The reaction takes as follows:



## Segment 4(Chemical Kinetics)

**Molecularity:** The number of molecules (or ions, atoms) that collide in an elementary reaction to form products.

- Always a **whole number** (1, 2, or 3).

**Example:**



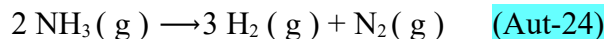
**Rate Law:** The mathematical expression that relates reaction rate to the concentration of reactants:

$$\text{Rate} = k[\text{A}]^m[\text{B}]^n$$

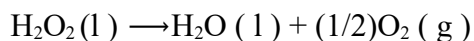
**Order of Reaction:** The sum of the powers of concentrations of reactants in the rate law:

$$\text{Order} = m + n$$

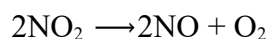
**Zero-order Reaction:** The reaction whose rate is independent of the concentration of a reactant, so that changing its concentration has no effect on the speed of the reaction is called zero-order reactions.



**First-order Reaction:** The reaction whose rate is proportional to first power of concentration of one reactant is called the first order reaction.

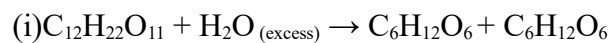


**Second-order Reaction:** The reaction whose rate is proportional to second power of concentration of one reactant or two reactants is called second order reaction.

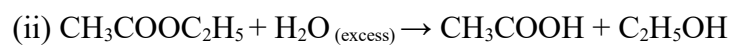


### Pseudo unimolecular reaction (Aut-24)

Definition: It is a reaction which seems to be second degree but actually a first order reaction. One of the reactant is in excess which makes its importance negligible and to follow the first order rate is called, "Pseudo Unimolecular Reaction". Example:



$$\text{Rate} = K [\text{C}_{12}\text{H}_{22}\text{O}_{11}]$$

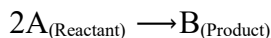


$$\text{Rate} = K [\text{CH}_3\text{COOC}_2\text{H}_5]$$

## Difference between Molecularity and Order of Reaction (Aut-24)

| Molecularity                                                                   | Order of Reaction                                                  |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------|
| No. of molecules colliding in an elementary step.                              | Sum of concentration powers in rate law.                           |
| Theoretical concept.                                                           | Experimental concept.                                              |
| Always whole number (1,2,3).                                                   | Can be zero, fractional, or whole.                                 |
| Mechanism of the reaction.                                                     | Experimental rate law.                                             |
| Does not change with temperature or pressure.                                  | May change under different conditions because rate law can change. |
| $\text{NO}_2 + \text{CO} \rightarrow \text{NO} + \text{CO}_2$ (Molecularity=2) | For the same reaction, order may differ                            |

### Deduce integrated equation for second order reaction.



initially,  $t = 0$ ,  $a \text{ moleL}^{-1} \longrightarrow 0$

after time,  $t = t$ ,  $(a-x) \text{ moleL}^{-1} \longrightarrow x$

If the initial concentration of A be  $a \text{ mol L}^{-1}$  and if in time “ $t$ ” the concentration of A will be  $(a-x) \text{ mol/L}$ . So according to law of mass action at any time  $t$  the rate of reaction will be:

$$\frac{dx}{dt} \propto (a-x)^2$$

$$\Rightarrow \frac{dx}{dt} = k (a-x)^2$$

$$\Rightarrow \frac{dx}{(a-x)^2} = k dt$$

$$\Rightarrow \int \frac{dx}{(a-x)^2} = \int k dt$$

$$\Rightarrow \frac{1}{a-x} = kt + C \dots \dots \dots (i)$$

When,  $t=0$ ,  $x=0$  then from eq<sup>n</sup> (i) we get,

$$\frac{1}{a} = 0 + C$$

$$\Rightarrow C = \frac{1}{a} \dots \dots \dots (ii)$$

Putting the value of C in eq<sup>n</sup> (i) we get,

$$\frac{1}{a-x} = kt + \frac{1}{a}$$

$$\Rightarrow kt + \frac{1}{a} = \frac{1}{a-x}$$

$$\Rightarrow kt = \frac{1}{a-x} - \frac{1}{a}$$

$$\Rightarrow kt = \frac{a-a+x}{a(a-x)}$$

$$\Rightarrow kt = \frac{x}{a(a-x)}$$

$$\Rightarrow k = \frac{1}{t} \cdot \frac{x}{a(a-x)}$$

## **Collision Theory of Chemical Reaction**

Collision theory was successfully applied to explain the bimolecular reactions in gaseous phase, but later it was found to be applicable also to be the reaction in solution. For reaction to take place :

- The first condition is- The Particles (atom, ion or molecules) taking part in a reaction must collide with one another.
- The second condition is- The collisions must take place with a minimum amount of energy.
- The third condition is- The collisions must take place with definite orientation. Arrhenius equation is in fact the mathematical expression of these conditions

## Segment 5 (Types and Properties of Solution)

### Show that Lowering of Vapour Pressure & Elevation of Boiling Point are Colligative Properties

**Colligative properties** depend on the **number of solute particles**, not their nature.

- Adding a non-volatile solute to a solvent:
- Vapour pressure decreases → because solvent molecules are fewer at the surface (Raoult's Law).
- To boil, vapour pressure = external pressure → **higher temperature needed** → **boiling point elevation**.
- So both depend on **moles of solute**, proving they are colligative properties.

### Derive Raoult's Law for Relative Lowering of Vapour Pressure & Roles of Emulsion

The vapour pressure of the solution is therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction.

That is,  $p_s \propto \frac{N}{n+N}$   
Where,  $N$  = moles of solvent and  
 $n$  = moles of solute.

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Or,  $p_s = K \frac{N}{n+N}$  ..... (1)  
 $K$  being proportionality factor.  
In case of pure solvent,  $n=0$  and hence  
mole fraction of solvent =  $\frac{N}{n+N}$   
 $= \frac{N}{0+N} = 1$

Now, from eq<sup>n</sup> (1), the vapour pressure,  $p = K$

Therefore, the eq<sup>n</sup> (1) assumes the form:

$$p_s = p \frac{N}{n+N}$$
$$\Rightarrow \frac{p_s}{p} = \frac{N}{n+N}$$
$$\Rightarrow 1 - \frac{p_s}{p} = 1 - \frac{N}{n+N}$$
$$\therefore \frac{p - p_s}{p} = \frac{n}{n+N}$$

this is Raoult's Law.

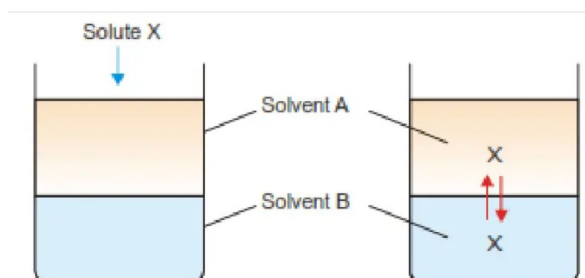
### Roles of Emulsion:

- Milk is an emulsion (oil in water) – food industry.
- Pharmaceutical creams – drug delivery.
- Emulsified fuels – energy efficiency.
- Cosmetic products (lotions, creams).

## Nernst's Distribution Law – Statement, Limitations, Applications

**Distribution Law:** If we take two immiscible solvents A and B in a beaker, they form separate layers. When a solute X, which is soluble in both solvents is added it gets distributed between them.

Molecules of X will pass from solvent A to B and from solvent B to A. Finally a dynamic equilibrium is set up.



At equilibrium, the rate at which molecules of X pass from one solvent to the other is balanced.

Solvents A and B at constant temperature and X is in the same molecular conditions in both solvents.

$$\frac{\text{Concentration of X in A}}{\text{Concentration of X in B}} = K_D$$

If  $C_1$  denotes the concentration of the solute in solvent A and  $C_2$  denotes the concentration of solute in solvent B.  $K_D$  is called Distribution co-efficient. Then Nernst's Distribution Law can be expressed mathematically as:

$$\frac{C_1}{C_2} = K_D$$

### Limitations of Distribution Law:

- 1) The temperature is kept constant throughout the experiment.
- 2) The solution must be dilute.
- 3) The two solvents must be immiscible.
- 4) The concentration of the solute is noted after the equilibrium has been reached.

### Applications of Distribution Law:

- 1) The distribution law has been successfully applied to the study of association or dissociation of a solute in a solvent.
- 2) It can be used to determine the equilibrium constant  $K_c$  for a complex reaction.
- 3) It finds numerous applications in the process of extractions of a solute from solvent both in the laboratory & in the industry.
- 4) It can be used to confirmatory test for bromide & iodide.
- 5) It can be applied to purify the lubricating oil.

## **Define colligate properties and mention four colligative properties**

The word colligative is derived from Latin word “Colligatus”, that means “to bind together or going together”. The essential features of colligative properties is that they depend only on the numbers of solute particles present in solutions. Being closely related to each other through a common explanation, these have been grouped together under the class name colligative properties.

Therefore, those properties of the solution, which depend only upon the total number of molecules of solute per unit volume & not on its chemical nature are called Colligative Properties.

Four main colligative properties of solutions are:

- 1) Lowering of vapour pressure.
- 2) Elevation of boiling point.
- 3) Depression of freezing point.
- 4) Osmotic pressure.